

QUADBIT

TEAM NAME: QUADBIT

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The Problem Statement

The BRSR Compliance Burden

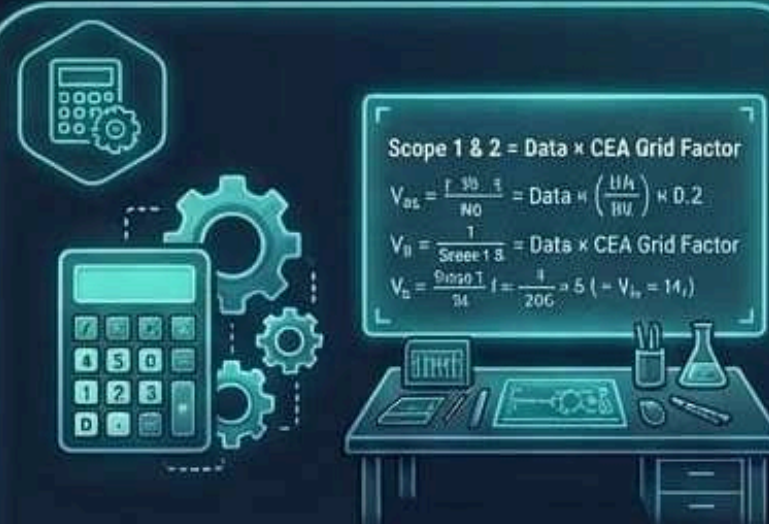
- **Regulatory Mandate:** SEBI requires top 1,000 listed entities to file complex BRSR Core disclosures by FY26-27.
- **Data Fragmentation:** Sustainability data (emissions, waste, diversity) is siloed across departments, making manual collection error-prone.
- **Complex Calculations:** Converting raw data into Scope 1 & 2 emissions using accurate, localized factors (like CEA grid factors) is highly technical.
- **Greenwashing Risks:** Inaccurate reporting leads to severe reputational damage and potential regulatory fines.

The BRSR Compliance Burden



REGULATORY MANDATE

SEBI requires complex BRSR Core disclosures for top 1,000 listed entities.



COMPLEX CALCULATIONS

Highly technical conversion of raw data using localized factors (like CEA).



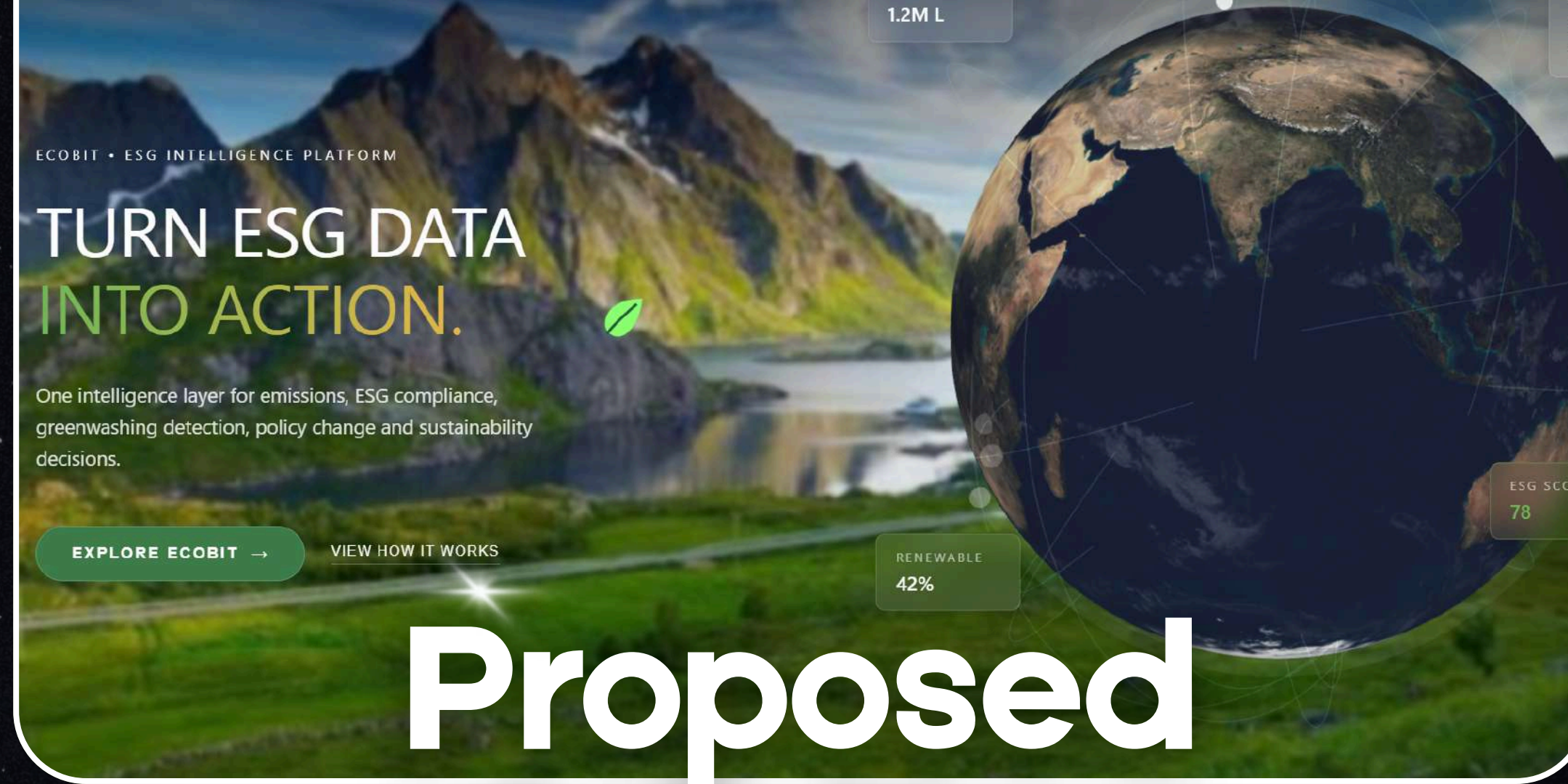
DATA FRAGMENTATION

Data siloed across departments, making manual collection error-prone.



GREENWASHING RISKS

Inaccurate reporting causes severe reputational damage & fines.



Proposed Solution

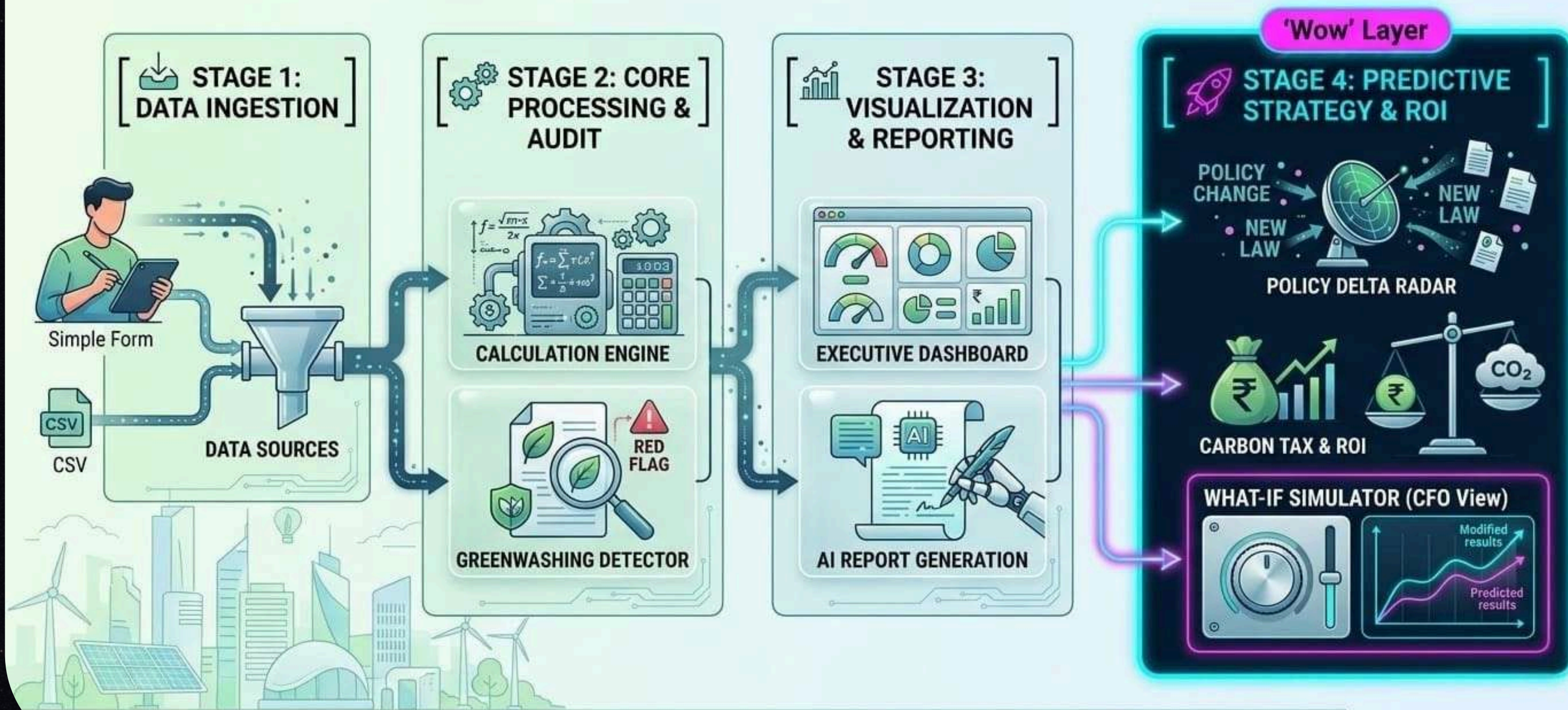
ECOBIT is a centralized, AI-powered platform that streamlines the entire ESG reporting lifecycle. We replace fragmented spreadsheets with a structured, automated workflow.

From seamless data ingestion via forms and CSVs to an automated calculation engine utilizing India-localized CEA emission factors, we ensure accuracy.

Our solution doesn't just calculate; it generates auditor-ready BRSR narrative reports, saving hundreds of manual hours.

FLOWCHART

SYSTEM ARCHITECTURE FLOWCHART: ECO-COMPLIANCE PLATFORM



Innovation & Uniqueness

The "Wow" Factor Features



Real-Time Carbon Tracking

Instantly monitor Scope 1, 2, and 3 emissions across all facilities using seamless, automated data collection.



AI-Powered Anomaly Detection

Proactively identify energy leaks, operational waste, and inefficiencies before they can impact your financial bottom line.



Automated ESG Compliance

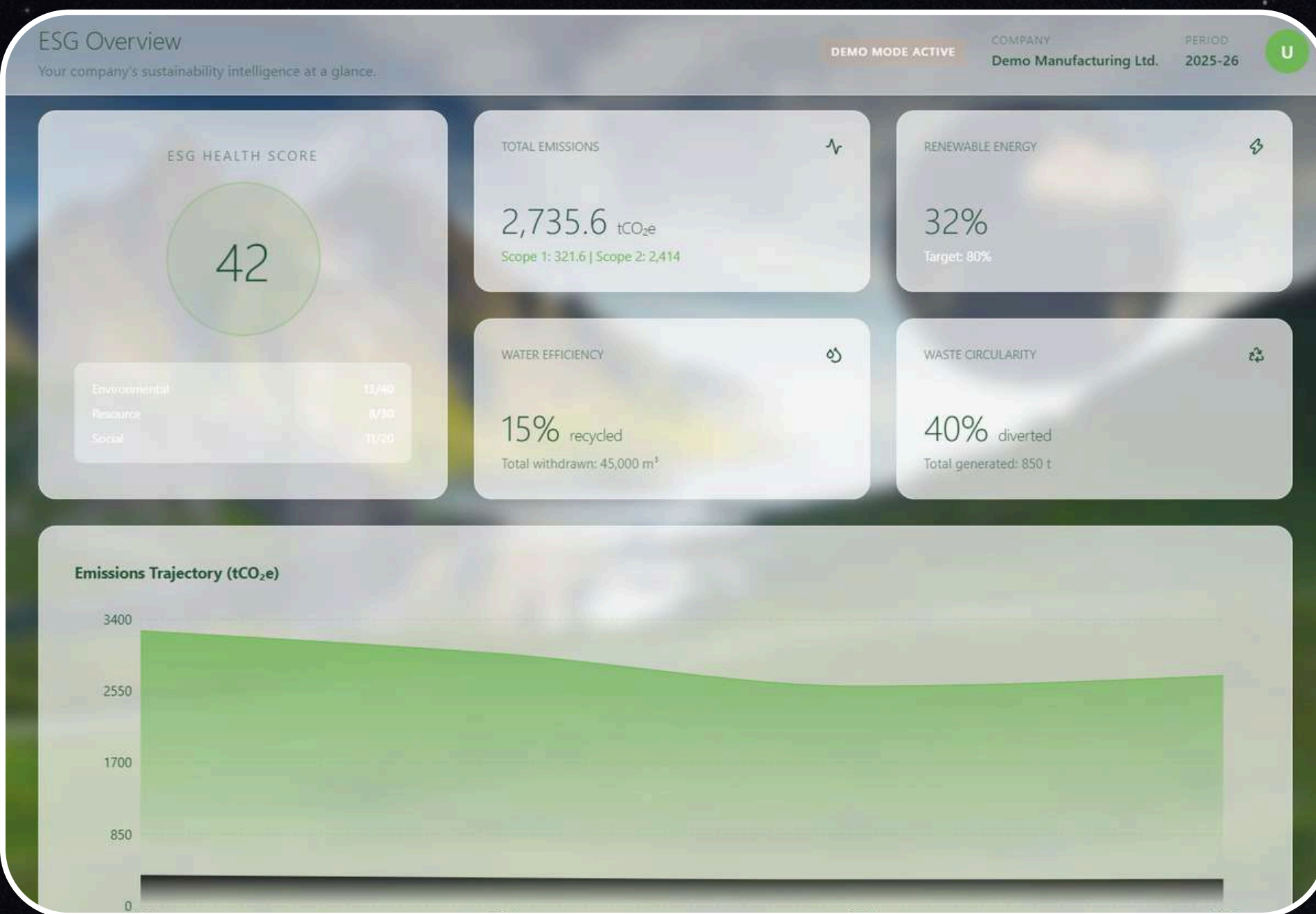
Generate error-free, one-click reports perfectly aligned with global sustainability frameworks, including BRSR and the GHG Protocol.



What-If Simulator

Simulate sustainability initiatives to accurately project carbon reductions, calculate annual energy cost savings, and determine exact ROI payback periods.

Key Features (MVP)



- **Data Input Hub:** Simple forms + CSV upload handling 4–5 core KPIs (Energy, Water, Waste, Diversity) for rapid onboarding.
- **India-Localized Calc Engine:** Hardcoded CEA grid emission factors automatically converting raw inputs to Scope 1 & 2 emissions.
- **AI Report Generation:** LLM integration takes calculated KPIs + company context to draft a narrative BRSR-Core-style section instantly.
- **Executive Dashboard:** Real-time visualizations featuring emissions trends, KPI vs. Target tracking, and a compliance checklist status.



Who benefits from ECOBIT?

- **Chief Sustainability Officers (CSOs):** Gain a centralized command center for all ESG metrics and compliance tracking.
- **Chief Financial Officers (CFOs):** Use the What-If Simulator to visualize cost savings and ROI on sustainability investments.
- **Compliance Teams:** Automate data gathering and report generation, minimizing errors and audit risks.
- **Investors & Regulators:** Access standardized, reliable, and verifiable ESG intelligence.

AI Integration: Utilizing advanced LLMs (like OpenAI GPT-4 or similar) for the automated BRSR narrative generation and Policy Delta Radar analysis.

Backend & Engine: Python/Django or Node.js to power the Calculation Engine and handle API integrations robustly.

Technology Stack

Robust, Scalable, Intelligent

Database: PostgreSQL for structured KPI data and MongoDB for flexible document storage of policies and reports.

Frontend: React.js or Next.js for a responsive, fast-loading Executive Dashboard and data input interfaces.

Impact & Scalability

Efficiency Gains via Automation (Estimated)

Impact: Reduces reporting time by up to 70%, minimizes greenwashing risks, and translates compliance into a strategic financial advantage.

Scalability: Built on a cloud-native architecture, ECOBIT can scale from top 1000 Indian entities to smaller supply chain partners (Scope 3) globally.





- **Whatsapp-to-scope 3:** An AI-powered WhatsApp bot that extracts energy data from supplier utility bill photos to automatically calculate and report Scope 3 emissions for enterprise dashboards.
- **Supply Chain Integration:** Expanding data ingestion to track Scope 3 emissions automatically from vendors and partners.
- **Blockchain Verification:** Adding an immutable layer for audit trails to ensure absolute data integrity and trust for regulators.

Future Scope

Beyond Basic Compliance